

JEE MAIN REVISION MASTER: COMPLEX NUMBERS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	De Moivre's Theorem & Cube Roots of Unity	5 PYQs	Roots of Unity
Moderate	Modulus, Argument & Polar/Euler Form	4 PYQs	Complex Basics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
De Moivre's Theorem & Cube Roots of Unity	$\omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \omega^2 = \bar{\omega},$ $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$
Modulus, Argument & Polar/Euler Form	$z = x + iy = r e^{i\theta}, \quad z ^2 = z \bar{z}, \quad \text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2),$ $ z_1 - z_2 \leq z_1 \pm z_2 \leq z_1 + z_2 $

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 De Moivre's Theorem & Cube Roots of Unity (Moderate Yield — 5 PYQs)

Core Principle: Algebraic and geometric properties of cube roots of unity forming an equilateral triangle on the unit circle, and De Moivre powers.

■ **Caution / JEE Pitfall:** $1 + \omega^n + \omega^{2n} = 3$ if n is a multiple of 3, and equals 0 if n is NOT a multiple of 3.

■ **Speed Trick:** Factorization identity: $x^3 - 1 = (x - 1)(x - \omega)(x - \omega^2)$, and $x^2 + x + 1 = (x - \omega)(x - \omega^2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\sum_{n=1}^4 (-1)^{2n} \cdot i^{2n} = \dots$ " [Ans: C]
- [2024 Shift] Q#3 (MCQ): "The real value of ' α ' for which $(1 - i \sin \alpha) / (1 + 2i \sin \alpha)$ is purely real is..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "If $3x + i(4x - y) = 6 - i$ where x and y are real numbers, then the values of x and y are respectively,..." [Ans: N/A]

3.2 Modulus, Argument & Polar/Euler Form (Moderate Yield — 4 PYQs)

Core Principle: Complex plane geometry, modulus properties, principal argument in $(-\pi, \pi]$, and triangle inequalities.

■ **Caution / JEE Pitfall:** Principal argument $\text{Arg}(z)$ must lie strictly in $(-\pi, \pi]$; if z is in the second quadrant, $\text{Arg}(z) = \pi - \tan^{-1}|y/x|$.

■ **Speed Trick:** Geometric interpretation: $|z - z_0| = r$ represents a circle centered at z_0 with radius r .

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "If z_1 and z_2 are two non-zero complex numbers, then which of the following is not true?..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "The modulus of the complex number $\frac{(1+i)^2(1+3i)(2-6i)(2-2i)}{(1-i)^2}$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If $((1+i)/(1-i))^x = 1$, then..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\sum_{n=1}^4 (-1)^{2n} \cdot i^{2n} =$$

- (A) 2
- (B) -i
- (C) 0 **[Correct]**
- (D) i

Correct Answer: C

Solution Analysis: $(-1)^{2n}$ is always equal to 1, because the power $2n$ is even. So, $(-1)^{2n} \cdot i^{2n} = 1 \cdot i^{2n} = i^{2n}$. Now use $i^2 = -1$. $i^{2n} = (i^2)^n = (-1)^n$. So the given sum becomes $\sum_{n=1}^4 (-1)^n = 1(-1)^1 + 1(-1)^2 + 1(-1)^3 + 1(-1)^4 = -1 + 1 - 1 + 1 = 0$. So the correct option is 0 Option C.

Benchmark Problem 2: KCET 2023 (Q#4)

The modulus of the complex number $\frac{(1+i)^2(1+3i)(2-6i)(2-2i)}{(1-i)^2(1-3i)(2+6i)(2+2i)}$ is

- (A) $\frac{2\sqrt{2}}{3}$
- (B) $\frac{1\sqrt{2}}{3}$
- (C) $\frac{\sqrt{2}}{4}$
- (D) $\frac{4\sqrt{2}}{3}$

Correct Answer: None

[illegible]